

Kali Linux System Information and Network Testing

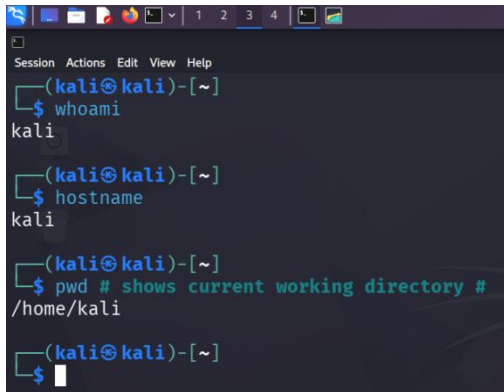
by

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Part 1

Step 1: System Information- Kali Linux Virtual Machine



```
(kali㉿kali)-[~]
$ whoami
kali

(kali㉿kali)-[~]
$ hostname
kali

(kali㉿kali)-[~]
$ pwd # shows current working directory #
/home/kali

(kali㉿kali)-[~]
$
```

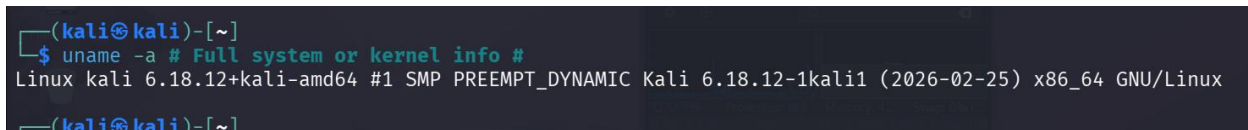
The username is kali.

Machine host name is kali

Current working directory is /home/kali

Step 2

- Operating Systems and Hardware Information.

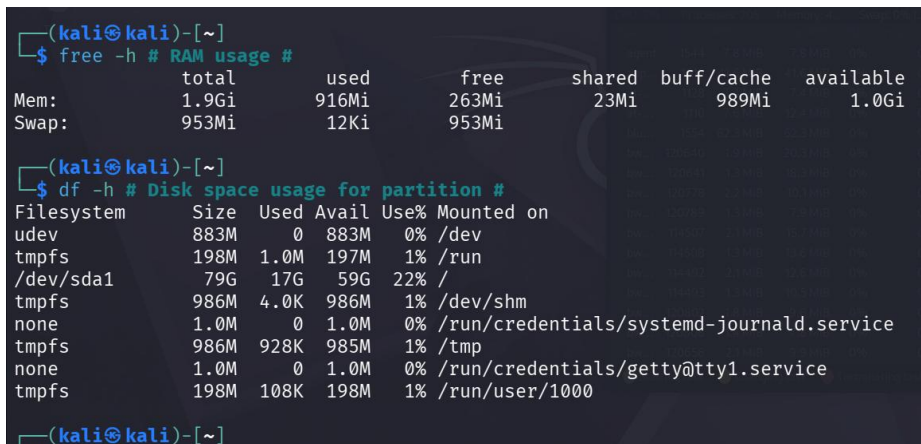


```
(kali㉿kali)-[~]
$ uname -a # Full system or kernel info #
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux

(kali㉿kali)-[~]
```

The version of the linux kernel of this kali machine is 6.18.12 -amd64

- RAM usage and Disk Space Available



```
(kali㉿kali)-[~]
$ free -h # RAM usage #
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	916Mi	263Mi	23Mi	989Mi	1.0Gi
Swap:	953Mi	12Ki	953Mi			

```
(kali㉿kali)-[~]
$ df -h # Disk space usage for partition #
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	883M	0	883M	0%	/dev
tmpfs	198M	1.0M	197M	1%	/run
/dev/sda1	79G	17G	59G	22%	/
tmpfs	986M	4.0K	986M	1%	/dev/shm
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	986M	928K	985M	1%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	198M	108K	198M	1%	/run/user/1000

```
(kali㉿kali)-[~]
```

The total RAM of this virtual machine is 1.9 gigabytes.

The disk space available in the main partition is 59 gigabytes.

Part 2

Step 1: Basic network connectivity test

```
(kali㉿kali)-[~]  
$ ping -c 2 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=47.4 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=26.1 ms  
  
— 8.8.8.8 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 26.081/36.723/47.366/10.642 ms  
  
(kali㉿kali)-[~]
```

Average round-trip time is **36.723 ms** and there is no packet loss

Step 2

- Internet connectivity and DNS resolution

```
(kali㉿kali)-[~]  
$ ping -c 4 google.com  
PING google.com (142.251.216.110) 56(84) bytes of data.  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=1 ttl=117 time=59.5 ms  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=2 ttl=117 time=255 ms  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=3 ttl=117 time=343 ms  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=4 ttl=117 time=343 ms  
  
— google.com ping statistics —  
4 packets transmitted, 3 received, 25% packet loss, time 3009ms  
rtt min/avg/max/mdev = 59.481/219.056/342.822/118.412 ms  
  
(kali㉿kali)-[~]  
$
```

Resolved IP address of Google is **142.251.216.110**

Average Round Trip Time (rtt)

The average round trip time (rtt) in pinging google.com is **219.056 ms** and it is different from, and way more than the average rtt in pinging 8.8.8.8 directly.

The possible reasons are below:

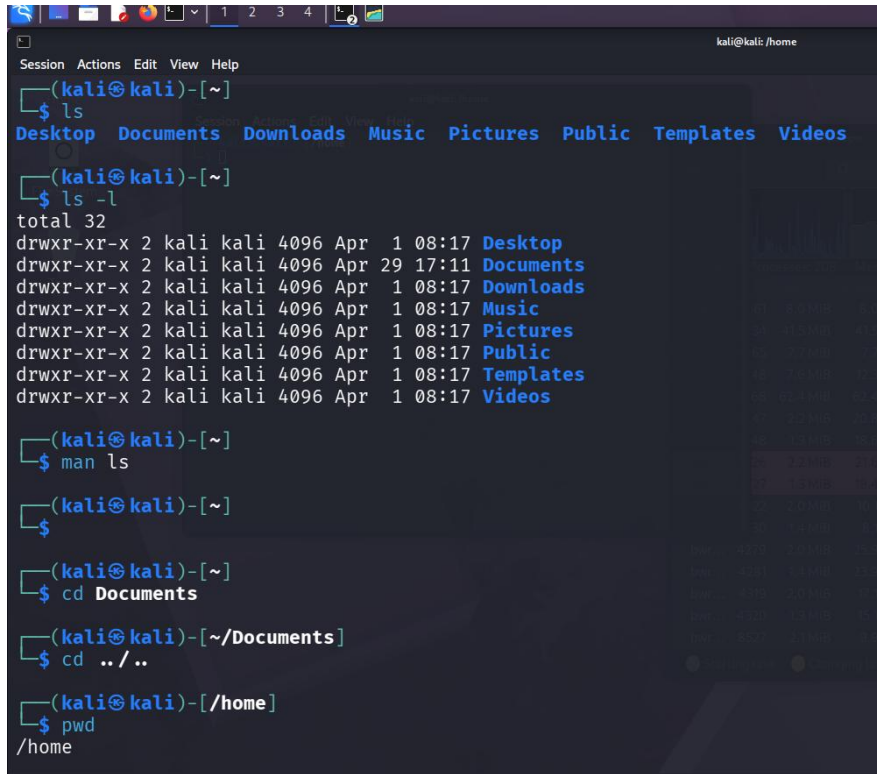
- **Different servers:** 8.8.8.8 is google public DNS server while google.com resolves to a google web server/CDN IP which is not 8.8.8.8. So, as the packets are sent to different servers, they will take different paths and will have different latencies.
- **DNS Lookup Time:** The first ping to Google.com will require a DNS resolution of Google.com IP address. This will have a DNS overhead and add a few ms on the first packet only, while 8.8.8.8 will skip that step as it is already an IP.
- **Packet count:** the packet count `-c`, affects RTT. As I pinged 8.8.8.8 with `-c 2`, the first packet tends to be slower due to ARP request and the second packet skips ARP because it has been caught. Meanwhile a ping on google.com with `-c 4` tend to result in 1 slow packet and 3 fast packets, making its average RTT slower.

What is `-c 4` ?

`-c 4` means 'count is 4'. It prompts ping to send 4 ICMP echo request packets then stop.

Other Deliverables:

- Listing and changing directories, and exploring man page.



```
(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

(kali㉿kali)-[~]
└─$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 29 17:11 Documents
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Downloads
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Music
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Pictures
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Public
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Templates
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Videos

(kali㉿kali)-[~]
└─$ man ls

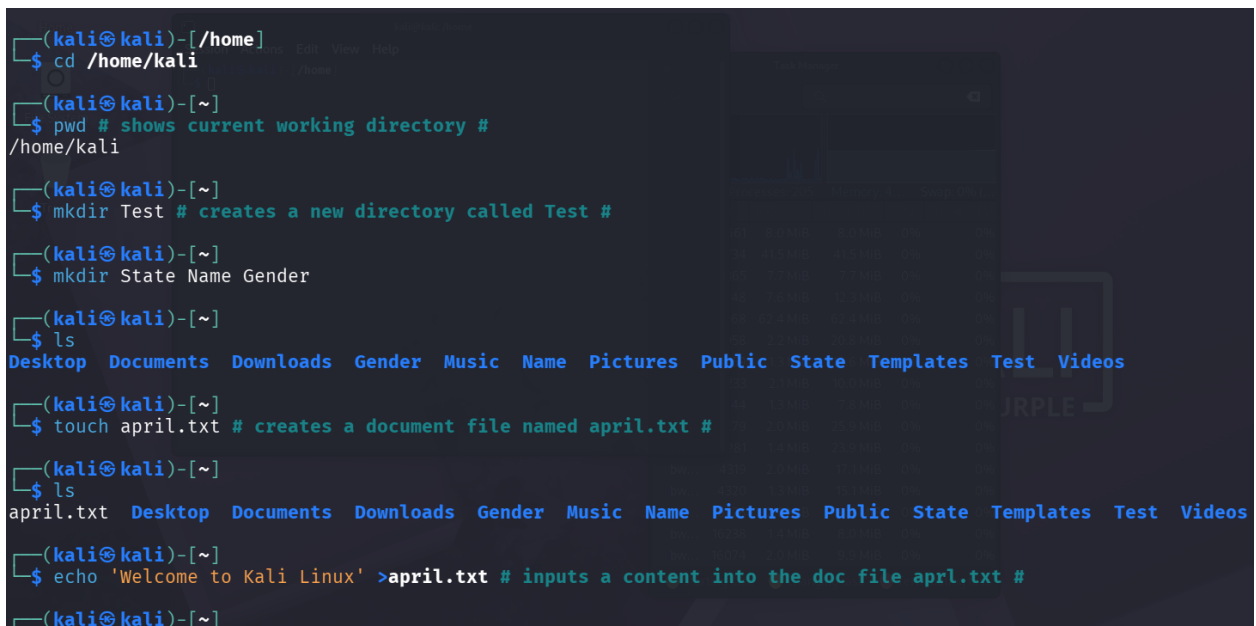
(kali㉿kali)-[~]
└─$

(kali㉿kali)-[~]
└─$ cd Documents

(kali㉿kali)-[~/Documents]
└─$ cd ../../

(kali㉿kali)-[/home]
└─$ pwd
/home
```

- Creating files and directories, and inputting content into a file.



```
(kali㉿kali)-[/home]
└─$ cd /home/kali

(kali㉿kali)-[~]
└─$ pwd # shows current working directory #
/home/kali

(kali㉿kali)-[~]
└─$ mkdir Test # creates a new directory called Test #

(kali㉿kali)-[~]
└─$ mkdir State Name Gender

(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos

(kali㉿kali)-[~]
└─$ touch april.txt # creates a document file named april.txt #

(kali㉿kali)-[~]
└─$ ls
april.txt  Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos

(kali㉿kali)-[~]
└─$ echo 'Welcome to Kali Linux' >april.txt # inputs a content into the doc file aprl.txt #

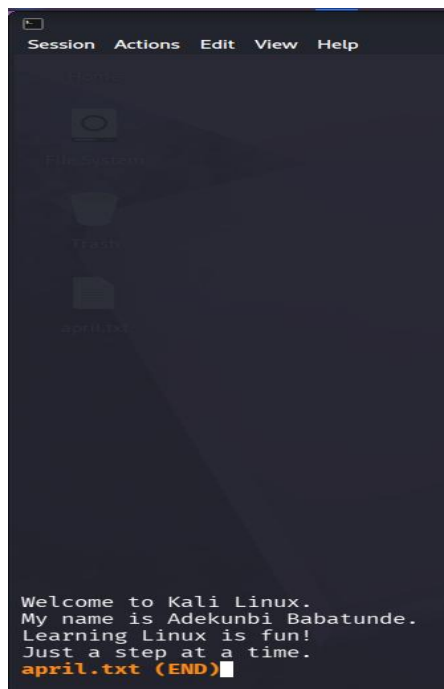
(kali㉿kali)-[~]
```

Displaying, copying and editing file content.

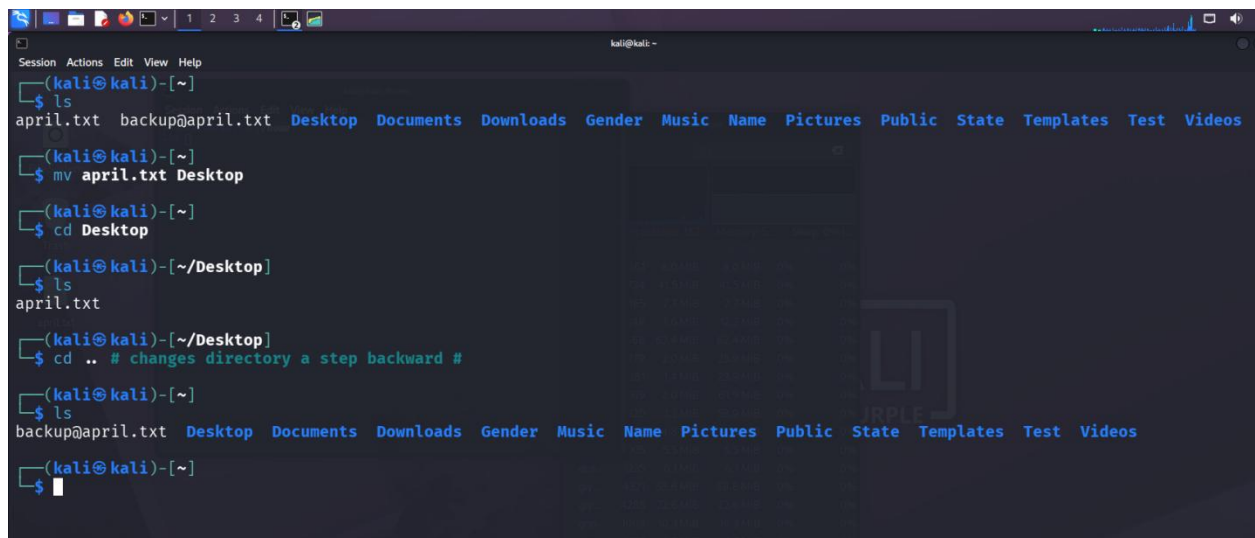
```
kali@kali: ~  
$ cat april.txt  
Welcome to Kali Linux  
  
(kali@kali)-[~]  
$ nano april.txt # Edits the content of april.txt #  
  
(kali@kali)-[~]  
$ cat april.txt # displays the content of the doc file #  
Welcome to Kali Linux  
I am Adekunbi Babatunde.  
Learning Linux is fun!  
Just take a step at a time.  
  
(kali@kali)-[~]  
$ cp april.txt backup@april.txt # makes a copy of april.txt into backup@april.txt #  
  
(kali@kali)-[~]  
$ cat backup@april.txt  
Welcome to Kali Linux  
I am Adekunbi Babatunde.  
Learning Linux is fun!  
Just take a step at a time.  
  
(kali@kali)-[~]  
$ ls  
april.txt  backup@april.txt  Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos  
  
(kali@kali)-[~]  
$
```

- Exploring less command

```
(kali@kali)-[~]  
$ cat april.txt  
Welcome to Kali Linux.  
My name is Adekunbi Babatunde.  
Learning Linux is fun!  
Just a step at a time.  
  
(kali@kali)-[~]  
$ less april.txt # to display a pagenated view #
```



- Changing and moving directories.



- Removing files and directories.

```

kali@kali ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ls
backup@april.txt Desktop Documents Downloads Gender Music Name Pictures Public State Templates Test Videos

(kali@kali)-[~]
$ rm backup@april.txt # removes backup@april.txt file #

(kali@kali)-[~]
$ rmdir Name Test # removes Name and Test directories #

(kali@kali)-[~]
$ ls
Desktop Documents Downloads Gender Music Pictures Public State Templates Videos

(kali@kali)-[~]
$

```

- Installing updates and advance package tools in Kali

```

(kali@kali)-[~]
$ sudo apt update
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [908 kB]

```

```

(kali@kali)-[~]
$ sudo apt install curl # Installs curl #
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmap python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio
gir1.2-vte-2.91 python3-filedepot python3-spinners
greenbone-feed-sync python3-filteralchemy python3-sqlalchemy-schemadisplay
gvmd python3-flask-celery-helper python3-standard-imghdr
gvmd-common python3-flask-classful python3-status
imagemagick python3-flask-kvsession python3-syslog-rfc5424-formatter
imagemagick-7.q16 python3-flask-limiter python3-tables
libapache2-mod-php python3-flask-login python3-tables-lib

```



```
(kali㉿kali)-[~]
└─$ curl --version # confirms curl installation #
curl 8.19.0 (x86_64-pc-linux-gnu) libcurl/8.19.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh
2/1.11.1 nghttp2/1.68.0 ngtcp2/1.21.0 nghttp3/1.12.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-03-11, security patched: 8.19.0-3
```

```
(kali㉿kali)-[~]
└─$ sudo apt remove curl # removes curl #
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmap python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio
gir1.2-vte-2.91 python3-filedepot python3-spinners
greenbone-feed-sync python3-filteralchemy python3-sqlalchemy-schemadisplay
gvmd python3-flask-celery-helper python3-standard-imghdr
```

```
Use 'sudo apt autoremove' to remove them.
```

```
REMOVING:
curl

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1114
  Freed space: 513 kB

Continue? [Y/n] y
(Reading database... 392514 files and directories currently installed.)
Removing curl (8.19.0-3)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...
```

```
(kali㉿kali)-[~]
└─$
```

```
(kali㉿kali)-[~]
└─$ curl --version # confirms curl installation removal #
Command 'curl' not found, but can be installed with:
sudo apt install curl
Do you want to install it? (N/y)
```